

Three genuinely novel solutions

1. Fifth Airy moment

For

$$a_5 = \int_0^\infty \text{Ai}(x)^5 dx,$$

define the Kampé de Fériet function

$$\mathcal{H}(x, y) = \sum_{j, k \geq 0} \frac{(\frac{2}{3})_j (1)_j (\frac{1}{2})_k (\frac{4}{3})_{j+k}}{(\frac{4}{3})_j (\frac{11}{6})_{j+k}} \frac{x^j y^k}{j! k!}.$$

Then

$$a_5 = \frac{F_1\left(1; \frac{1}{3}, \frac{1}{2}; \frac{3}{2}; \frac{1}{16}, \frac{1}{4}\right)}{24\pi^2 3^{2/3} \Gamma(\frac{2}{3})} - \frac{\mathcal{H}(\frac{1}{16}, \frac{1}{4})}{3^{1/3} 48^{4/3} \pi^{3/2} \Gamma(\frac{1}{3}) \Gamma(\frac{11}{6})}.$$

Proof

Introduce the quartic Mellin transform

$$M(s) = \int_0^\infty x^{s-1} \text{Ai}(x)^4 dx.$$

Its known exact value is

$$M(s) = \frac{\Gamma(s)}{48^{(s+2)/3} \pi^{3/2} \Gamma((2s+7)/6)} {}_2F_1\left(\frac{s+2}{3}, \frac{1}{2}; \frac{2s+7}{6}; \frac{1}{4}\right).$$

The remaining Airy factor has the expansion

$$\text{Ai}(x) = \text{Ai}(0)_0 F_1\left(\frac{2}{3}; \frac{x^3}{9}\right) + \text{Ai}'(0) x_0 F_1\left(\frac{4}{3}; \frac{x^3}{9}\right).$$

Termwise integration is justified by Tonelli's theorem applied to the two nonnegative component series, together with Airy's exponential decay. This produces sums involving $M(3j+1)$ and $M(3j+2)$. Gamma triPLICATION and expansion of the remaining ${}_2F_1$ resum these respectively to the Appell F_1 value and $\mathcal{H}$ above.

This is an exact transformation of the improper integral into fixed named two-variable hypergeometric constants. It uses no fitting, quadrature, or accuracy-dependent truncation.

2. Equal-power TE+TM spherical-mode quality factor

Let

$$L = n(n+1), \quad c = \sqrt{L}, \quad m = n+1,$$

and define

$$\hat{a}_0 = 1, \quad \hat{a}_{k+1} = \hat{a}_k \frac{(n+k+1)(n-k)(2k+1)}{2(k+1)L}.$$

Set

$$D(y) = \sum_{k=0}^n \frac{\hat{a}_k y^{k+1}}{k+1}, \quad P(v) = (1+v)^m \left[2 - D\left(\frac{1}{1+v}\right) \right].$$

Let A be the companion matrix of the monic polynomial

$$\frac{(-1)^m P(-u)}{[v^m]P(v)},$$

and, crucially, use the principal matrix square root:

$$S = cA_{\text{principal}}^{1/2}, \quad K = \pi \left(\text{tr } S - \frac{(2n+1)c}{2} \right).$$

For $x \leq c$, with $y = L/x^2$,

$$Q_n^{\text{TE+TM}}(x) = K + x \left(\frac{D(y)}{2} - 1 \right).$$

For $x > c$, with $z = \sqrt{x^2 - L}$,

$$\begin{aligned} Q_n^{\text{TE+TM}}(x) = & K + x \left(\frac{D(y)}{2} - 1 \right) \\ & - 2 \text{tr} [S \{ \arctan(zS^{-1}) - zS^{-1} \}] \\ & + (2n+1)c \{ \arctan(z/c) - z/c \}. \end{aligned}$$

The reciprocal-argument form in the implementation is algebraically identical and is used only to avoid large- x cancellation.

Proof

For $u(\rho) = \rho h_n^{(2)}(\rho)$, the half-integer Hankel expansion makes $|u|^2$ an exact terminating polynomial in L/ρ^2 , with coefficients $\hat{a}_k$. The Riccati–Bessel equation and Hankel Wronskian reduce the reflection weight above cutoff to

$$w = \frac{1}{2} - \frac{\sqrt{1 - L/\rho^2}}{2 - D(L/\rho^2)}.$$

The first half of the integrand has an elementary finite-polynomial primitive. After $v = (\rho^2 - L)/L$, the remaining rational term is expressed through $P'(v)/P(v)$. Encoding P with its companion matrix turns this logarithmic derivative into a trace, whose integral is exactly the displayed matrix-arctangent expression. Continuity at cutoff and the condition $Q(\infty) = 0$ determine K .

The result is exact for every $n \geq 1$ and $x > 0$. Its matrix dimension is fixed by n , rather than by a requested numerical accuracy, so it involves no quadrature, numerical root finding, or approximation-dependent truncation.

3. Non-resonant TM/TE spherical-mode quality factor

Let

$$N = n(n+1), \quad m = n+1, \quad \alpha = \frac{1}{2} + i\sqrt{N - \frac{1}{4}}, \quad \beta = \bar{\alpha},$$

and define the terminating hypergeometric polynomial

$$H_n(x) = {}_6F_3\left(\begin{matrix} \frac{1}{2}, \alpha+1, \beta+1, -n, n+1, 1 \\ 1-\alpha, 1-\beta, 2 \end{matrix}; -\frac{1}{x^2}\right).$$

For $\ell = 0, \dots, m-1$, put

$$r_\ell = m - \ell, \quad a_\ell = \binom{m}{r_\ell} N^{r_\ell} {}_3F_0\left(\begin{matrix} -r_\ell, -\frac{1}{2}, n \\ - \end{matrix}; -\frac{1}{N}\right).$$

Let $C \in \mathbb{C}^{2m \times 2m}$ be the Frobenius companion matrix defined by

$$C_{j+1,j} = 1, \quad C_{2\ell+1,2m} = -a_\ell,$$

with all other entries zero, using one-based indices. Define

$$z = \begin{cases} 0, & x \leq \sqrt{N}, \\ \sqrt{x^2 - N}, & x > \sqrt{N}, \end{cases} \quad \theta = \begin{cases} -\pi/2, & z = 0, \\ -\arctan(\sqrt{N}/z), & z > 0, \end{cases}$$

and use the principal matrix logarithm

$$\mathcal{L} = \begin{cases} \log(-C), & z = 0, \\ \log(zI - C), & 0 < z \leq 1, \\ \log(I - C/z), & z > 1. \end{cases}$$

Then

$$Q_n^{\text{TM}}(x) = Q_n^{\text{TE}}(x) = z - x + \frac{N}{2x} \operatorname{Re} H_n(x) + (4m - 2)\sqrt{N} \theta + \operatorname{Re} \operatorname{tr} \left[\left(C - \frac{C^3}{N} \right) \mathcal{L} \right].$$

Proof

For $\psi(\rho) = \rho h_n^{(2)}(\rho)$,

$$R(\rho) = |\psi(\rho)|^2 = {}_3F_0\left(-n, n+1, \frac{1}{2}; -; -\rho^{-2}\right),$$

which terminates exactly. The Riccati–Bessel equation gives

$$|\psi'|^2 + \frac{N}{\rho^2} R = R + \frac{1}{2} R''.$$

The Pochhammer identity

$$\frac{(\alpha+1)_k(\beta+1)_k}{(1-\alpha)_k(1-\beta)_k} = \frac{N+k(k+1)}{N}$$

then shows that $-\rho + NH_n(\rho)/(2\rho)$ is an exact primitive of the unweighted integrand. Above cutoff, the Wronskian reduces the reflection weight to a rational expression. Multiplying its denominator by $\rho^{2n+2}/2$ produces exactly the characteristic polynomial $\det(zI - C)$. Jacobi’s log-determinant identity and Newton identities integrate the remaining correction into the matrix-log trace. Positivity of the physical denominator validates the principal logarithm branches; cutoff continuity and decay at infinity complete the proof.

The identity is exact and uses only terminating hypergeometric functions and a finite matrix of dimension $2(n+1)$. The submitted fixed-100-digit implementation can lose accuracy through catastrophic cancellation for astronomically large x ; adaptive working precision or a far-field branch is needed for uniform numerical stability. This is a conditioning issue, not a defect in the identity.